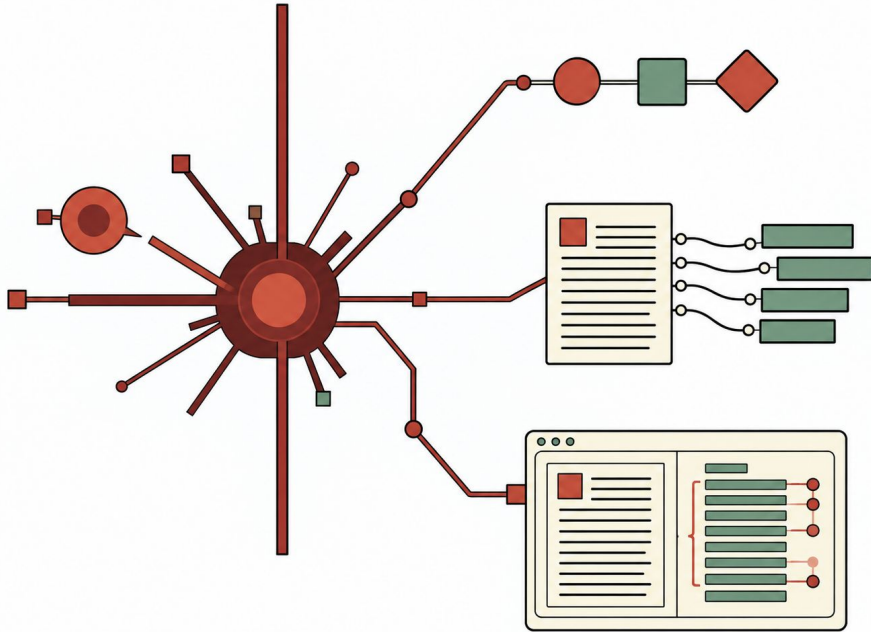




Knowledge, Context and Agentic Engineering for Research Data Workflows & Digital Editions

CLARIAH-AT Summer School 2026
Machine Learning for Digital Scholarly Editions.
25.09.2026

[Slides](#)

Dr. Christopher Pollin MA MA
Digital Humanities Craft OG
www.dhcraft.org | office@dhcraft.org



Applied Generative AI for Research Data Workflows

YouTube <https://www.youtube.com/@DigitalHumanitiesCraft>

Slides <https://chpollin.github.io/GM-DH>

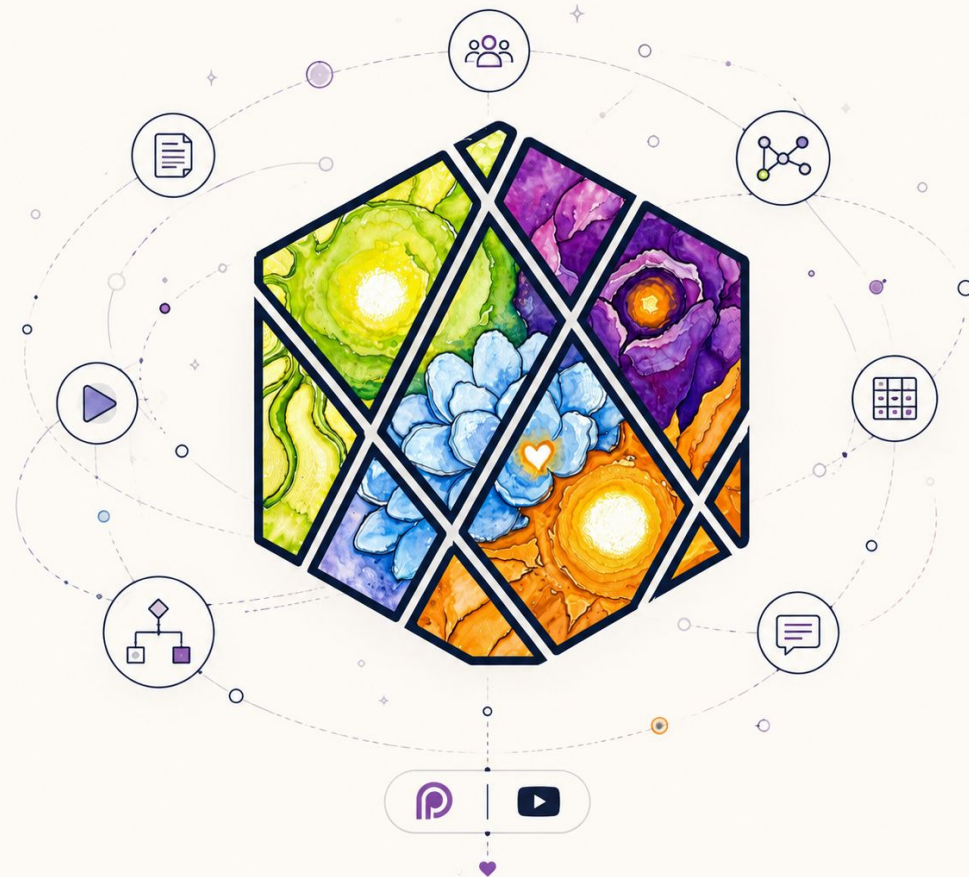
Blog <https://dhcraft.org/excellence/blog>

Zotero Library <https://www.zotero.org/groups/5319178/agki-dh>

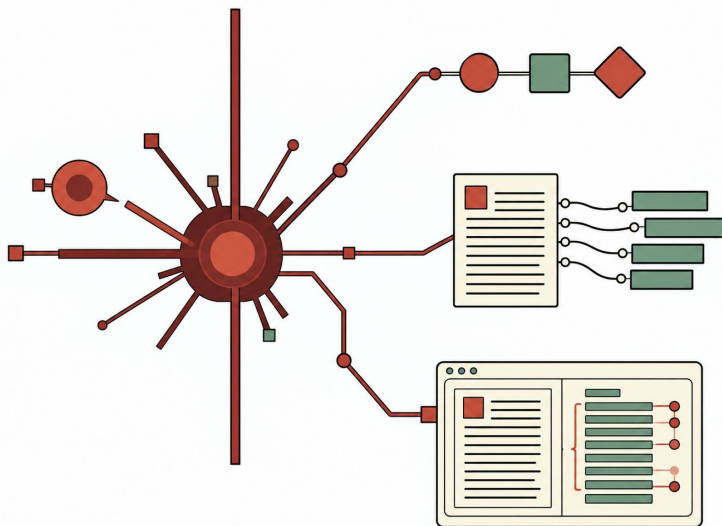
Patreon <https://www.patreon.com/c/DigitalHumanitiesCraft>

<https://chpollin.github.io>

<https://dhcraft.org>



Frontier LLMs amplify computer-based research asymmetrically, with early signs of recursive self-improvement



Growing capabilities

AI agents can handle increasingly demanding computational tasks.

METR: longer task horizons

FrontierMath: advanced mathematics

Jagged capabilities

Strength in one task does not ensure reliability in another.

SimpleBench: everyday reasoning

ARC-AGI-3: adaptation to unfamiliar tasks

- Opus 5.5: [How Close Are We to Automated AI Research?](#)
- [My Playlist](#)

Asymmetric amplification

Research gains vary with the task, expertise, and project context.

Asymmetric Amplification

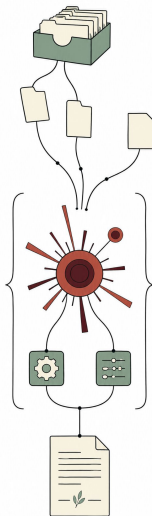
Learning Objectives

Use **Promptotyping** and an **AI harness** to build and evaluate an **agentic pipeline for digital scholarly editions**.

- Organise **project knowledge, context, and tools** to guide the pipeline.
- Prepare **source material** and define **extraction requirements**.
- Implement and run the pipeline; check **output structure and accuracy against the sources**.
- Compare findings across letters and support conclusions with **source evidence**.

Core Concepts

- **Prompt engineering:** Design and test task instructions.
- **Knowledge engineering:** Maintain explicit, revisable project knowledge.
- **Context engineering:** Select and organise the information the model needs for a task.
- **Agentic engineering:** Design and test workflows in which AI agents use tools and execution feedback to carry out tasks.
- **AI agent:** A system that selects actions towards a goal.
- **AI harness:** Software that manages the agent's execution.



Which models and AI tools do you use?

Which hosted frontier services do you use, and who has paid access?



[ChatGPT](#)
OpenAI



[Claude](#)
Anthropic



[Gemini](#)
Google



[Grok](#)
xAI



[Mistral](#)
France



[Llama](#)
Meta



[DeepSeek](#)
China



[Qwen](#)
Alibaba, China



[Kimi](#)
Moonshot AI, China

Who uses open source or open weight models, and who runs models locally?

Local access



[Ollama](#)



[LM Studio](#)



[Claude Code](#)



[Codex](#)



[Gemini CLI](#)



[Mistral Vibe](#)

Which AI harness do you use?



[Qwen Code](#)



[Kimi Code](#)



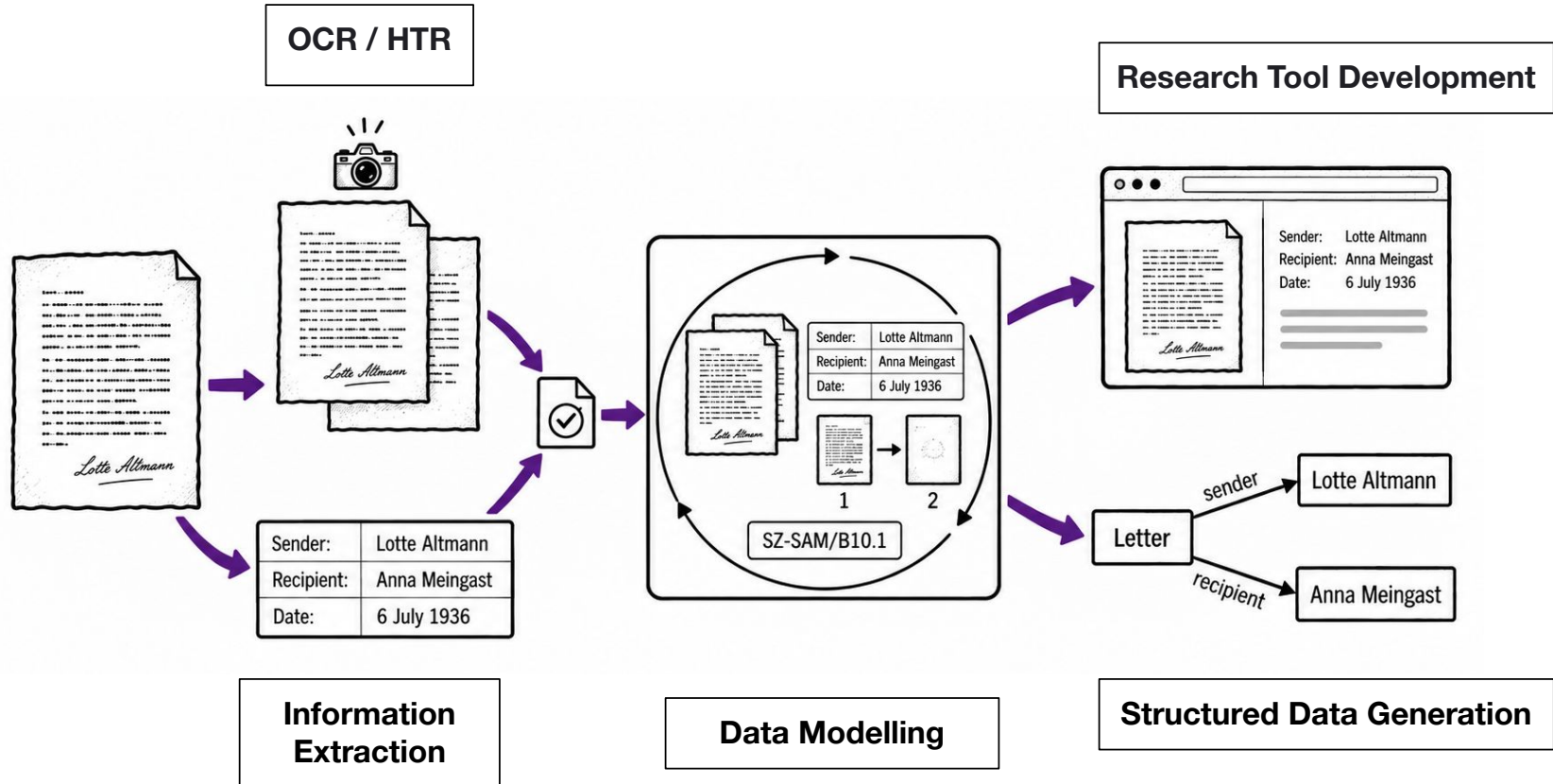
[Pi](#)



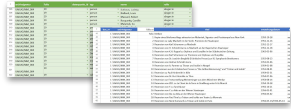
[Cursor](#)

Frontier LLM Landscape: <https://artificialanalysis.ai>

Research Data Workflow and LLMs



Research Data



Archiv	Indizes	Mobilität	Matrix	Kosmos	Zeitraum	Partitur	Korb
...	...	...	...	...	...	...	...

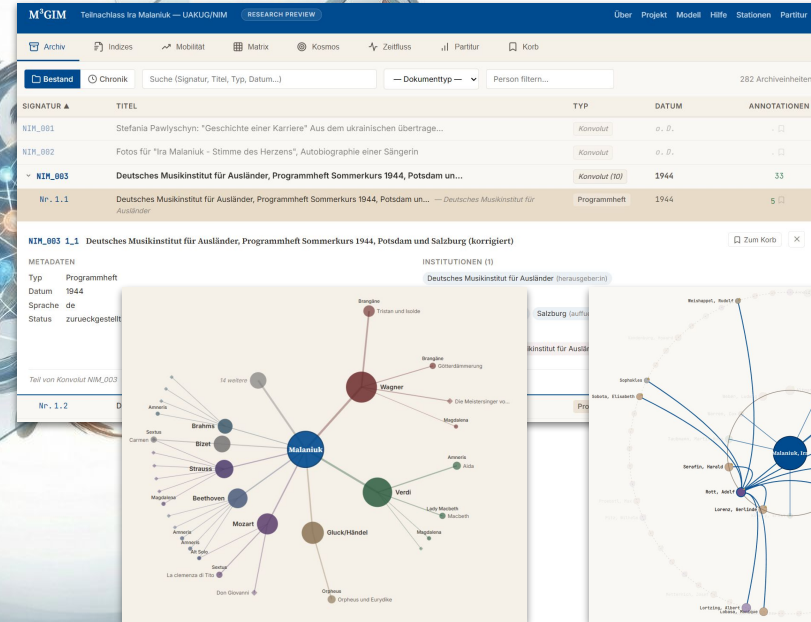
Promptotyping

RQ1. How did singers shape Graz's music and theatre culture, and how did mobility contribute to professionalisation and networks?

RQ2. Which narrative and aesthetic structures were influenced by migration, and how did they contribute to transforming opera?

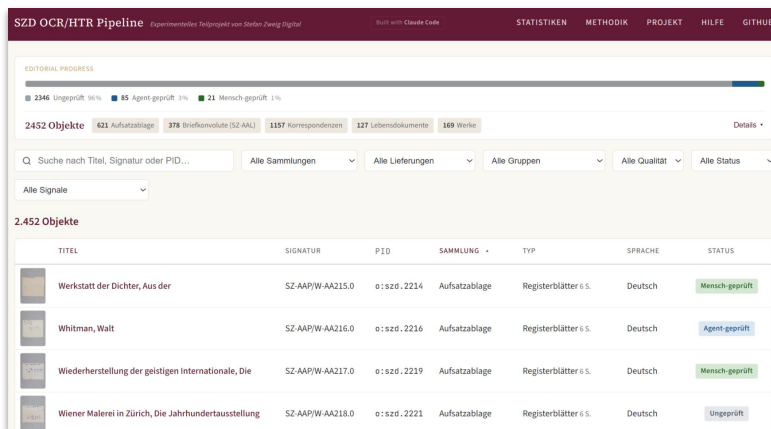
(Frontier-)LLM & Context Engineering

<https://dhcraft.org/Promptotyping>
<https://github.com/DigitalHumanitiesCraft/Promptotyping>



Research Artefacts (e.g. tools, workflows, models)

Agentic Edition Pipeline and ‘Epistemic’ Infrastructure



SZD OCR/HTR Pipeline Experimentelles Teilprojekt von Stefan Zwarg Digital Built with Claude Code STATISTIKEN METHODIK PROJEKT HILFE GITHUB

EDITORIAL PROGRESS

■ 2346 Ungeprüft: 96% ■ 85 Agent-geprüft: 3% ■ 21 Mensch-geprüft: 1%

2452 Objekte 621 Aufsatzablage 378 Briefkomplete (SZ-AAL) 1157 Korrespondenzen 127 Lebensdokumente 169 Werke Details

Q Suche nach Titel, Signatur oder PID... Alle Sammlungen Alle Lieferungen Alle Gruppen Alle Qualität Alle Status

Alle Signale

2.452 Objekte

TITEL	SIGNATUR	PID	SAMMLUNG	TYP	SPRACHE	STATUS
 Werkstatt der Dichter, Aus der	SZ-AAP/W-AA215.0	o:szd.2214	Aufsatzablage	Registerblätter 6 S.	Deutsch	Mensch-geprüft
 Whitman, Walt	SZ-AAP/W-AA216.0	o:szd.2216	Aufsatzablage	Registerblätter 6 S.	Deutsch	Agent-geprüft
 Wiederherstellung der geistigen internationale, Die	SZ-AAP/W-AA217.0	o:szd.2219	Aufsatzablage	Registerblätter 6 S.	Deutsch	Mensch-geprüft
 Wiener Malerei in Zürich, Die Jahrhundertausstellung	SZ-AAP/W-AA218.0	o:szd.2221	Aufsatzablage	Registerblätter 6 S.	Deutsch	Ungeprüft

Manuscripts · Gemini 3.1 Flash Lite · HTR · curation

Frontend: <https://chpollin.github.io/szd-htr-ocr-pipeline>
Repository: <https://github.com/chpollin/szd-htr-ocr-pipeline>

Agentic Edition Pipeline Repo:
<https://github.com/DigitalHumanitiesCraft/agentic-edition-pipeline>

Repository: Versions research data, code and project knowledge, preserving the history of changes.

Workflows and pipelines: Connect OCR/HTR, layout analysis, TEI encoding and entity annotation through explicit data contracts.

Validation: Checks software behaviour, data structure and TEI schema conformance.

Verification: Research software developers check implementation and execution. Editors assess readings and interpretations against the sources.

Epistemic infrastructure: Connects tools, evidence and human responsibilities so that generated results can be traced, challenged and corrected, with unresolved uncertainty kept explicit.

Our Workshop Project: Build and Check a Digital Edition Workflow

Work with the supplied letters or your own source material. We will use Promptotyping to guide implementation through short knowledge documents that record requirements, decisions and findings.

1. Transcribe one page with a vision-language model and check the result.
2. Run transcription through the Gemini API, review a pilot page and approve further processing.
3. Build a local viewer with source images on the left and transcriptions on the right.
4. Optionally, generate TEI-XML using agreed encoding rules.
5. Optionally, explore topics and link a visualisation to supporting passages in the viewer.

Review each result before continuing. Record confirmed changes in the project knowledge.

Working with Your Own Data

Adapt the prompts to your source files, language, transcription rules and research question. Start with one page and check that the agent's assumptions match your material.

Hands-on: Initialise Your *Agentic Edition Pipeline* Project

- **Prepare your workspace**
 - Create a **Workshop** folder and open it in [Visual Studio Code](#).
 - Start [Claude Code](#) or your preferred AI harness in this folder.
- **Choose your settings**
 - Select an available model. Use a mode where you have enough **usage!**
 - Use **Medium reasoning effort**, where supported.
- **Add your data**
 - Download the [2 English-language letters from the Hugo Schuchardt Archive](#) and extract the package into **data/**.
 - Alternatively, place your own source materials there.
- **Initialise your project**
 - Paste the supplied setup prompt to initialise the [Agentic Edition Pipeline](#) using [Promptotyping](#).
 - Answer the agent's questions about your data, research goal and available tools.
 - Let the agent configure your workspace and create **knowledge/plan.md**.

Prepare the current folder for our digital edition workshop. It already contains a folder with starting data. Work directly in the current folder. Do not clone repositories.

Identify and inspect the starting-data folder. If its location is unclear, ask us. Preserve all supplied files and their paths.

Create:

- output/ Generated transcriptions, TEI-XML and analysis results
- scripts/ Processing code
- viewer/ Static image-text viewer
- knowledge/ Project knowledge that guides our work

Use the existing starting-data folder as the source directory.
Do not move or duplicate it.

Inspect its file structure and formats. Examine one suitable letter and any corresponding images, transcriptions or TEI. Summarise what is available, how the files relate and what requires clarification. Distinguish supplied texts from outputs we will generate.

Our workshop milestones are:

1. Transcribe one page with Gemini 3.5 Flash Lite, review it and approve further processing.
2. Build a local image-text viewer with OpenSeadragon.

Optional extensions, selected by us:

- Generate TEI-XML using agreed encoding rules.
- Explore topics and visualise their supporting passages.

After each milestone, present the results and check them against the agreed acceptance criteria. Wait for our explicit approval before starting the next milestone.

Create these concise Markdown files in knowledge/:

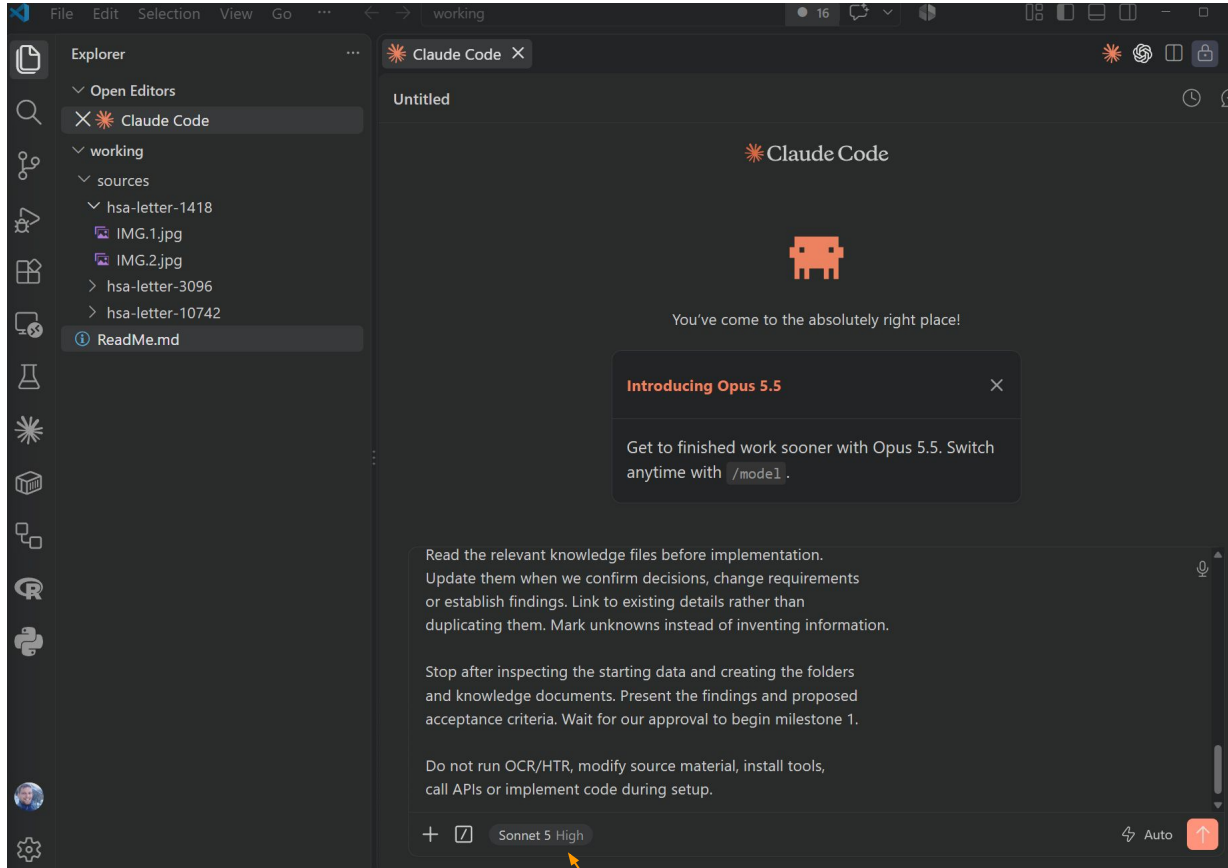
- data.md: Document the inspected source material, its paths, provenance, formats, relationships and known limitations.
- specification.md: Record the milestones and propose acceptance criteria for each. Mark these as pending our agreement.
- plan.md: Outline the two milestones with their inputs, checks and first executable action. Keep optional extensions separate until we select them.

Read the relevant knowledge files before implementation.

Update them when we confirm decisions, change requirements or establish findings. Link to existing details rather than duplicating them. Mark unknowns instead of inventing information.

Stop after inspecting the starting data and creating the folders and knowledge documents. Present the findings and proposed acceptance criteria. Wait for our approval to begin milestone 1.

Do not run OCR/HTR, modify source material, install tools, call APIs or implement code during setup.



Better use Medium effort here!

Initialisation

- New Claude Code Session in Visual Studio Code
- Your sources in a sources or data Folder
- Prompt that prepares the structure of the repo
- ... what is the AI agent doing on your machine?
- If you use a better model, as Sonnet 5, you will find everything is smoother and working better, but it is more expensive!

Hands-On: Transcribe a facsimile with a VLM

Markdown

Task

Context

Rules

Output

Create a diplomatic transcription of this facsimile

Context

This page comes from an English-language scholarly letter written in 1885.

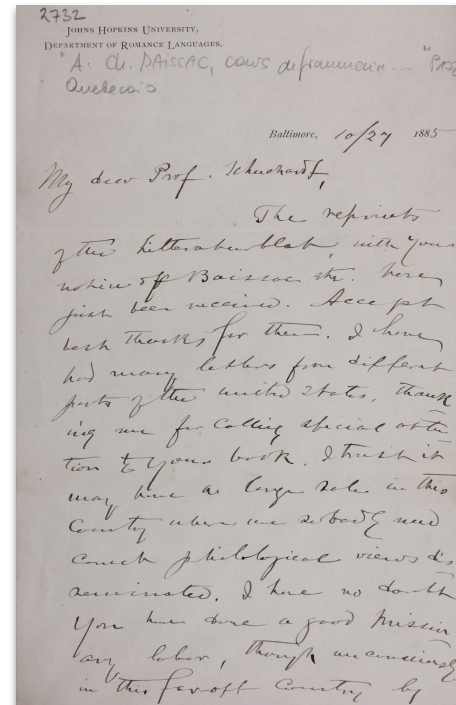
Rules

- * Transcribe printed and handwritten text
- * Preserve spelling, abbreviations, line breaks and field order
- * Mark uncertainty as word[?] and illegible text as [...]
- * Mark deletions as ~text~ and insertions as {text}
- * Ignore show-through
- * {{add here a new rule}}

Output

Return only the transcription in reading order

Prompt



Gemini 3.5 Flash Lite

Result from the ChatBot

Model • 21:16

2732

JOHNS HOPKINS UNIVERSITY,
DEPARTMENT OF ROMANCE LANGUAGES.

"A: A. BAISSAC, cours de grammaire -- P78
Québécois"

Baltimore, 10/27 1885

My dear Prof. Schuchardt,

The reprints
of the Litteraturblatt, with your
notice of Baissac etc. have
just been received. Accept
best thanks for them. I have
had many letters from different
parts of the United States, thank
ing me for calling special atten
tion to your book. I trust it
may have a large sale in this
country where we so badly need
sound philological views &c.
disseminated. I have no doubt
You have done a good mission
ary labor, though unconscious,
in this far-off country by



Google AI models may make mistakes, so double-check outp

2732

JOHNS HOPKINS UNIVERSITY,
DEPARTMENT OF ROMANCE LANGUAGES.

"A: A. BAISSAC, cours de grammaire -- P78
Québécois"

Baltimore, 10/27 1885

My dear Prof. Schuchardt,

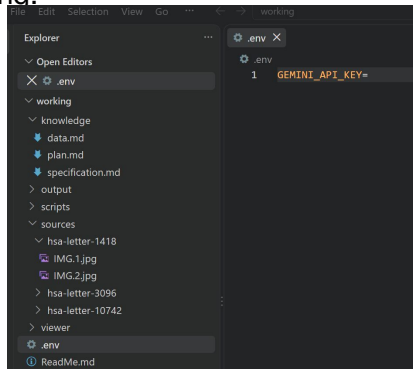
The reprints
of the Litteraturblatt, with your
notice of Baissac etc. have
just been received. Accept
best thanks for them. I have
had many letters from different
parts of the United States, thank
ing me for calling special atten
tion to your book. I trust it
may have a large sale in this
country where we so badly need
sound philological views &c.
disseminated. I have no doubt
You have done a good mission
ary labor, though unconscious,
in this far-off country by

Hands-On: Transcribe Your First Page with the Gemini API

I will provide a Gemini API key for this exercise.

1. Open or create `.env` in your project folder. Preserve any existing settings.
2. Add your key, replacing the placeholder:
`GEMINI_API_KEY=paste_your_key_here`
3. Save `.env` and add `.env` to your `.gitignore` file. Keep the key private and never paste it into the chat.
4. Paste your revised transcription prompt from the previous exercise into your AI harness together with the the implementation prompt on this slide. Review the first page before approving further processing.

- We let the model do a web search to better “understand” how to use Gemini 3.5 Lite.
- You can give it the prompt we already have as an example:
- If the first page worked, you can run it for all!



Read the project knowledge in `knowledge/` and locate the supplied images and our prepared transcription prompt.

Consult the official Gemini API documentation for image input, authentication and the exact identifier for Gemini 3.5 Flash Lite. Use only this model. If you cannot verify its identifier or access it, stop and report the issue.

Create a script in `scripts/` that sends one image and our prepared transcription prompt to the Gemini API.

Load `GEMINI_API_KEY` from `.env` without displaying or logging it. Ensure `.env` is excluded from Git. If the key is missing, stop and tell us how to add it.

Use the selected page, or the first page of one letter if none has been selected. If the transcription prompt is missing, ask us.

Keep existing TEI and transcriptions out of the API input. Preserve all source files.

Process only this page. Save the transcription as UTF-8 text in a separate run folder under `output/transcriptions/`. Save the prompt, exact model identifier and source path alongside it. Never overwrite earlier results.

Show the transcription and the source image or its local file link. Ask us to check missing text, incorrect readings, reading order and adherence to our transcription rules.

Wait for our feedback. Ask before repeating the API call. Process further pages only after our explicit approval. Update the relevant knowledge files briefly. Do not publish.

Hands-On: Build a Viewer to Check Your Transcriptions

Read knowledge/data.md and knowledge/specification.md.
Inspect the source images and generated transcriptions.

Build a static viewer in viewer/ using HTML, CSS and JavaScript.
Use OpenSeadragon for the image display, stored locally.

Requirements:

- Display the source image on the left and its transcription on the right.
- Support image zoom and pan.
- Provide page selection and previous/next navigation.
- Keep the image and transcription matched when changing pages.
- Preserve line breaks and display transcription notation literally, including uncertainty, deletions and insertions.
- Show which source image and transcription file are displayed.
- If several transcription runs exist, ask which to use.
- Report missing or ambiguous file matches rather than guessing.

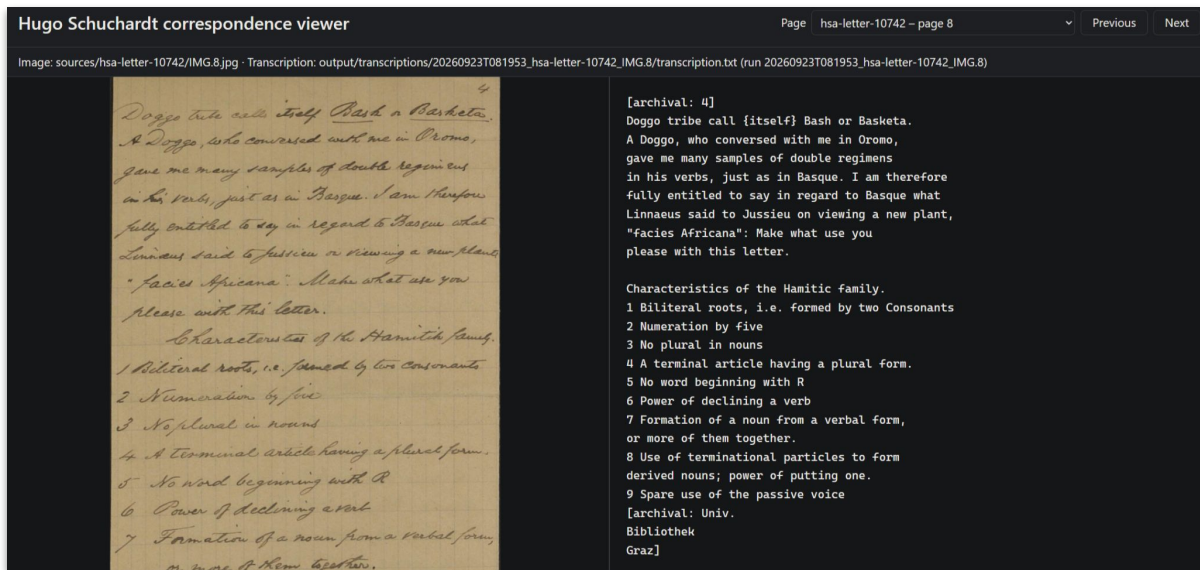
Keep the viewer read-only. Preserve all source files and transcriptions. Use local files without a backend or API calls.

Start a local static server and verify that the viewer loads, page navigation works and each image matches its transcription. Give us the local browser URL.

Update the existing knowledge files briefly with the confirmed file mappings, implementation and checks.

Ask us to compare a page with its transcription and report reading errors or changes needed in the viewer.
Stop and wait for feedback. Do not generate TEI or publish.

Hands-On: Review and Improve Your Viewer



Give feedback

Try the viewer. Describe a problem, how to reproduce it and what you expected. Share a screenshot or error message with the agent.

Add tests

Ask the agent to reproduce the problem and add a test that detects it. Fix the cause and rerun the test.

Refactor the code

Ask the agent to simplify unnecessarily complex or duplicated code while preserving working behaviour. Run the tests again.

Refine the project knowledge

Update confirmed requirements and decisions. Remove outdated or duplicated information and keep unresolved questions explicit.

Hands-On: Explore and Visualise Topics in Your Letters

Which topics connect or distinguish the two letters? Choose a research question, review the suggested topics and design a visualisation alongside your viewer. Each topic should lead to supporting passages and their source images.

Read the project knowledge, available transcriptions and existing viewer code.

Help us explore topics in the two letters. Ask which research question or interest should guide the exploration. If we have none, suggest directions grounded in the texts.

Propose a small set of topics with exact supporting passages, letter identifiers and page references. Treat the assignments as unreviewed suggestions. A passage may support several topics. Flag uncertain readings that could affect the interpretation.

Discuss the proposed topics with us and suggest suitable ways to visualise them alongside the image-text viewer.

Wait for our choice and corrections before implementation.

Extend the existing static website with the agreed visualisation:

- Let us select a topic and inspect its supporting passages.
- Link each passage to its corresponding page in the viewer.
- Verify file mappings and report missing links rather than guessing.
- Keep source files and transcriptions unchanged.
- Store topic assignments, supporting passages and review status separately from the transcriptions.

Use local files so the website works without API calls.

Test the topic links, page mappings and interactions.

Start the local server and give us the browser URL.

Ask us to review the findings against the sources and try the visualisation.

Update the relevant knowledge files with confirmed decisions.

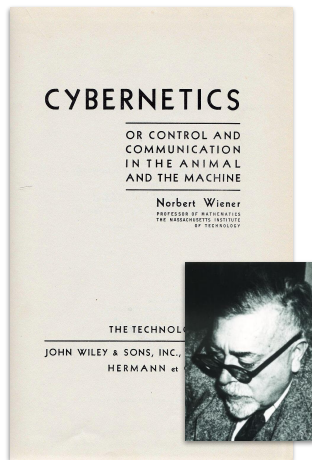
Stop and wait for feedback. Do not publish.

AI Agents Existed Long Before LLMs

An **AI agent** perceives an environment, selects actions, and pursues goals with some degree of autonomy.

- **Autonomy:** Acts without continuous supervision.
- **Reactivity:** Responds to environmental changes.
- **Proactiveness:** Initiates goal-directed actions.
- **Social ability:** Interacts with humans or other agents.

Wooldridge and Jennings (1995). Intelligent Agents: Theory and Practice.



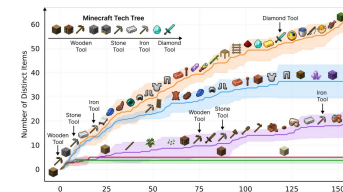
Norbert Wiener.
Cybernetics (1948).
Title page.



Multi-Agent
Hide and Seek
OpenAI (2019)
Watch video



AlphaGo
Google (2016)
Watch video

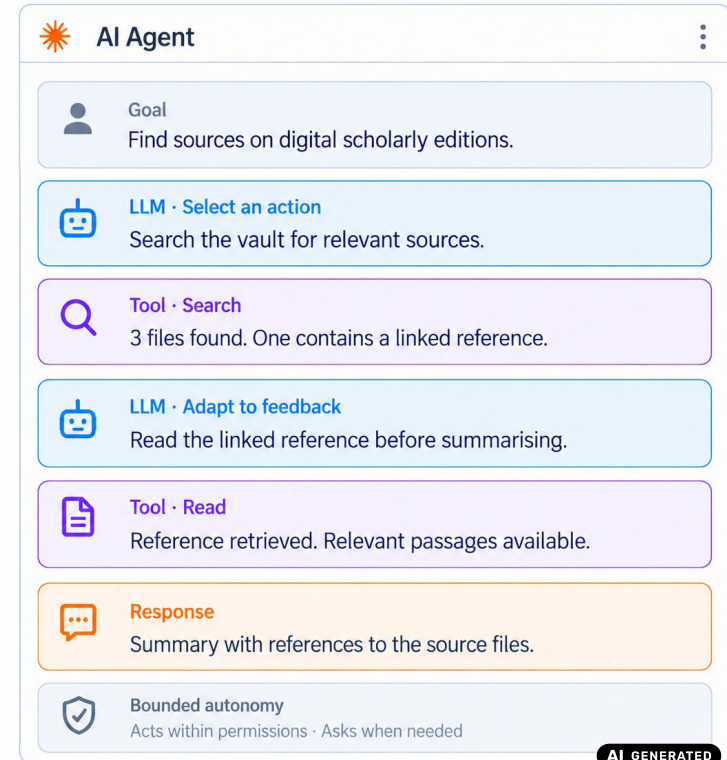


LLM-based
agents:
Voyager
Wang et al.
(2023)
Read paper

LLM-based AI Agent

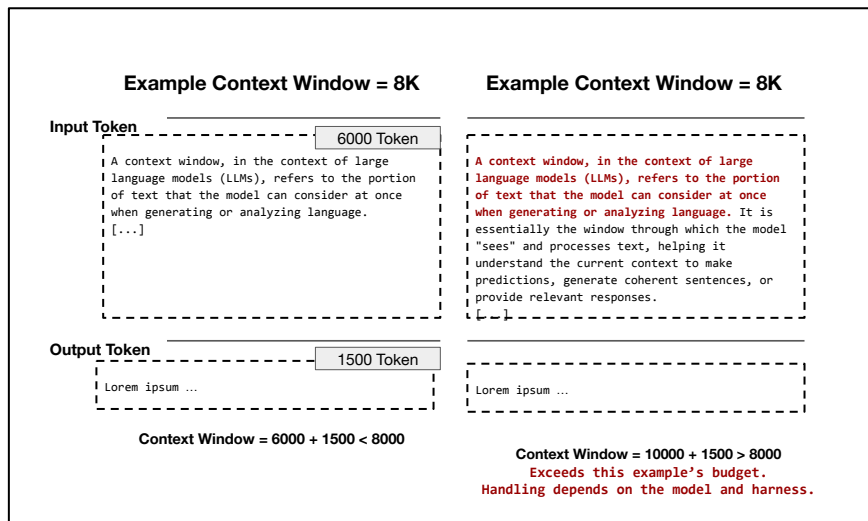
An AI agent is a system that uses a **language model to plan and select actions** (including **tool calls**) towards a **defined goal**. It incorporates **feedback from execution** to adjust subsequent steps, operating with bounded autonomy.

Sapkota, R., Roumeliotis, K. I., & Karkee, M. (2025). AI Agents vs. Agentic AI: A Conceptual Taxonomy, Applications and Challenges. *Information Fusion*, 126, 103599. <https://doi.org/10.1016/j.inffus.2025.103599>
Xu, W., Huang, C., Gao, S., & Shang, S. (2025). LLM-Based Agents for Tool Learning: A Survey. *Data Science and Engineering*, 10, 533–563. <https://link.springer.com/article/10.1007/s41019-025-00296-9>

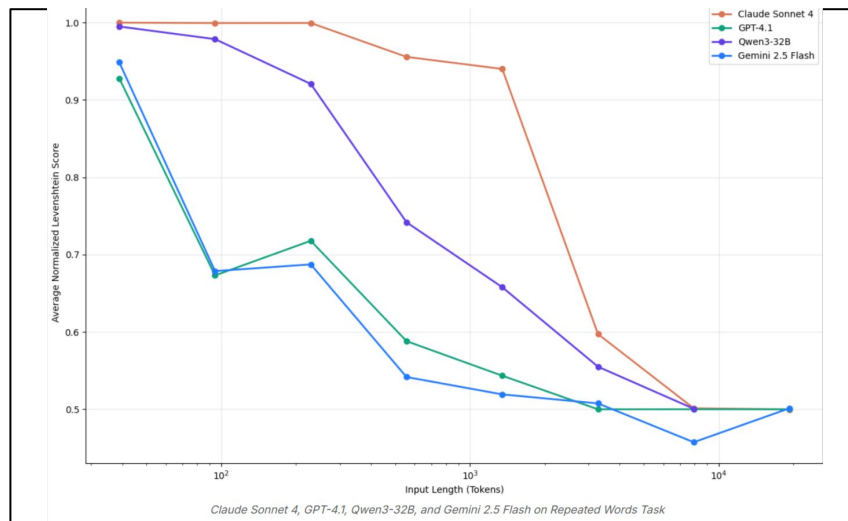


Context Engineering: Managing the Context Window

The **Context Window** is the model's finite working space, containing the input and previously generated tokens available at each generation step. Through *self-attention*, the model uses these tokens to predict the next token.



Context Rot describes how the model's ability to retrieve and use relevant information can decline as the number of tokens in the context window grows.



Markdown Makes Document Structure Explicit for LLMs

Markdown is a lightweight markup language that encodes document structure directly in human-readable plain text. For LLMs, it makes hierarchy and document elements explicit with little markup overhead.

- # Heading
 - ## heading 2
 - ### heading 3
- **Bold**
- - List item
- [Link](url)
- `code`
- ``` code blocks
- ...

<https://www.markdownguide.org>



```
## Transcript:
...
__Bestellzettel Bücherausgabe__[^source]

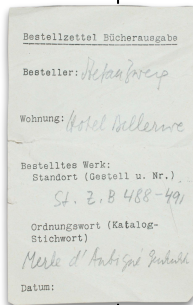
Besteller: Stefan Zweig
Wohnung: Hotel Bellevue

Bestelltes Werk:
Standort (Gestell u. Nr.)
St. Z. B 488-491

Ordnungswort (Katalog-
Stichwort)
Merle d'Aubigné Denkart[?]

Datum:
...

[^source]: lorem ipsum ...
```



```
<!-- Excerpt from word/document.xml -->

<w:document xmlns:w="...">
  <w:body>

    <w:p>
      <w:pPr>
        <w:pStyle w:val="Heading1"/>
      </w:pPr>
      <w:r>
        <w:rPr><w:i/></w:rPr>
        <w:t>Bestellzettel Bücherausgabe</w:t>
      </w:r>
    </w:p>

    ...
```



<https://platform.openai.com/tokenizer>

```
<!-- Excerpt from word/document.xml -->

<w:document xmlns:w="...">
  <w:body>

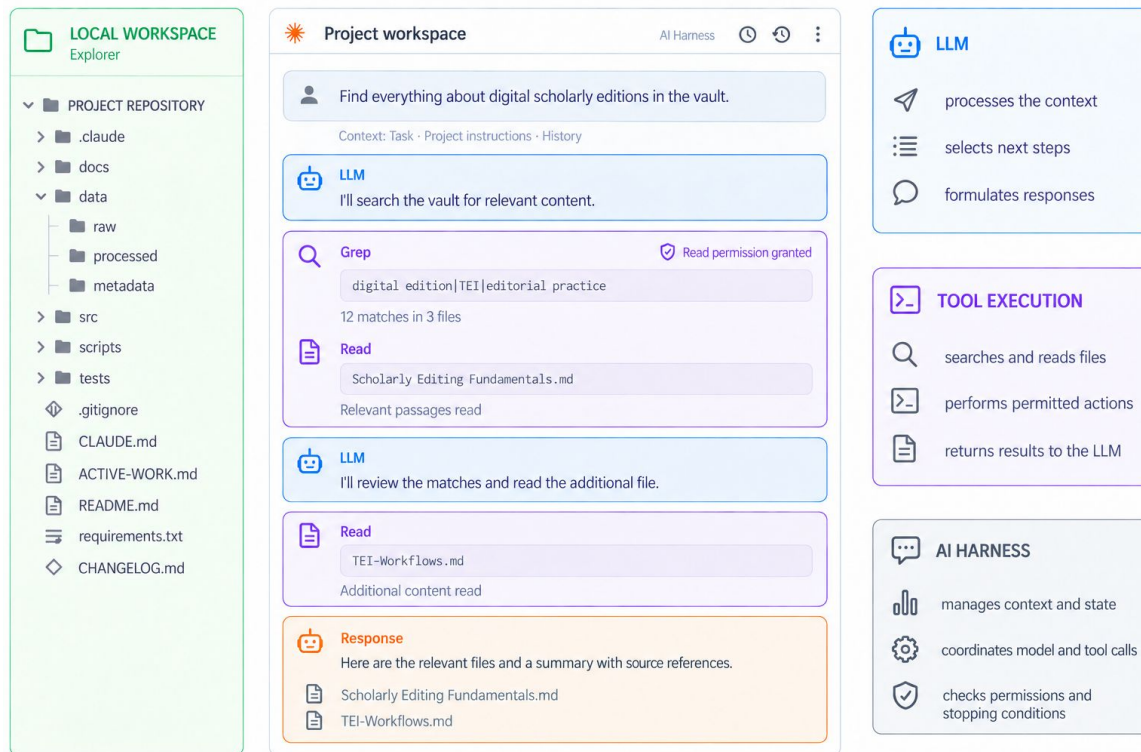
    <w:p>
      <w:pPr>
        <w:pStyle w:val="Heading1"/>
      </w:pPr>
      <w:r>
        <w:rPr><w:i/></w:rPr>
        <w:t>Bestellzettel Bücherausgabe</w:t>
      </w:r>
    </w:p>

    ...
```

AI Harness

An AI harness is the **software layer** that **runs the agent loop** around a model. It **manages context and state**, coordinates **model calls and tool execution**, and returns results to the model while enforcing permissions and stopping conditions.

- Claude Code. <https://claude.ai/code>
- OpenAI Codex. <https://chatgpt.com/codex>
- Cursor. <https://cursor.com>
- Windsurf. <https://windsurf.com>
- OpenCode. <https://opencode.ai>
- Pi. <https://pi.dev>
- Antigravity CLI. <https://antigravity.google>
- Qwen Code. <https://github.com/QwenLM/qwen-code>
- Cline. <https://cline.bot>
- GitHub Copilot. <https://github.com/features/copilot>
- Manus – <https://manus.im>
- ...



Knowledge Documents (*Promptotyping Documents*)

A **knowledge document** is a structured record of externalized knowledge, distilled from sources, data or experience for reuse by people and AI agents. Its claims remain traceable to evidence, with assumptions and uncertainties made explicit.

- **Dual-readable:** Makes knowledge understandable to people and usable as context by AI agents.
- **Compact:** Distils and synthesises information while preserving essential meaning, provenance and uncertainty.
- **Portable:** Uses open, plain-text formats such as Markdown for retrieval, linking, versioning and exchange.
- **Curatable:** Supports review, correction and enrichment as evidence and understanding develop.

```
▼ knowledge
  ↳ architecture.md
  ↳ data-model.md
  ↳ data.md
  ↳ design.md
  ↳ handoff.md
  ↳ INDEX.md
  ↳ journal-archive.md
  ↳ journal.md
  ↳ plan.md
  ↳ recording-guide.md
  ↳ research-framework.md
  ↳ specification.md
  ↳ testing.md
```

Templates for Knowledge Documents

data.md

```
# Data

## Subject
Selected documents about Ira Malaniuk.

## Sources
PDFs and available transcription drafts.
Record source IDs, filenames and provenance.

## Model
Documents → pages → text passages.
TEI links transcriptions to source pages.

## Limitations
Transcriptions require checking.
Place mentions do not confirm attendance.
```

research.md

```
# Research

## Question
Which places are
mentioned in the selected
documents, and in what
context?

## Method
Check transcriptions
against the PDFs.
Identify place mentions
and read them
in their surrounding
passages.

## Interpretation
Distinguish explicit
statements from
inferences. Record
unresolved readings.
```

specification.md

```
# Specification

## Requirements
Display transcription and source together.
Let users find passages mentioning a place.

## Scope
A small edition of the selected documents.
One main view and one core interaction.

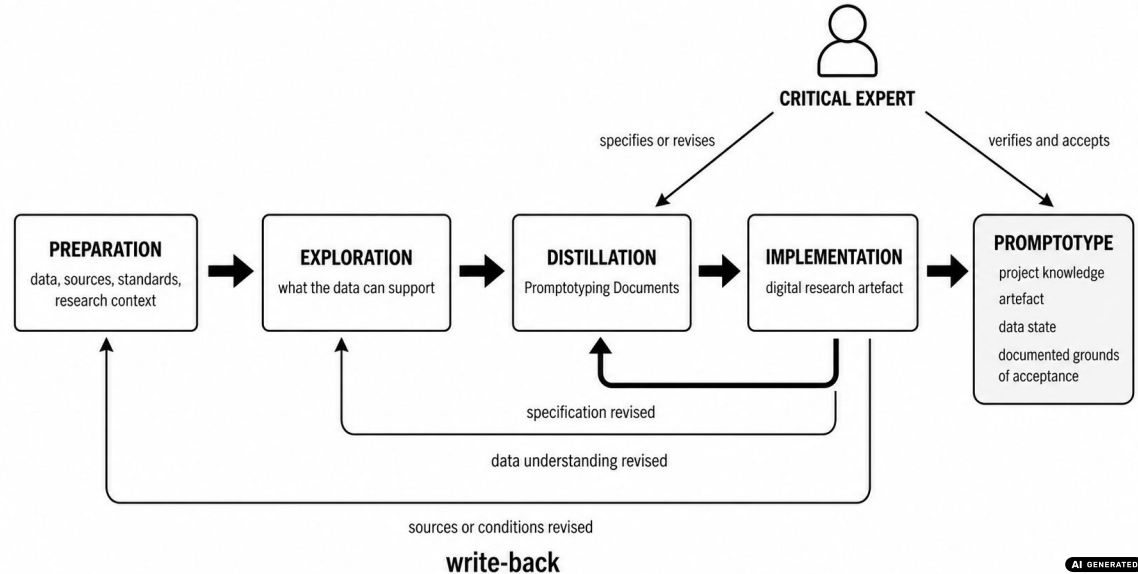
## Technical constraints
Use locally stored OpenSeadragon.
No frontend frameworks or build tools.

## Acceptance criteria
Each passage links to its source page.
Uncertain readings remain visible.
Source IDs and original files are preserved.
```

Promptotyping

Promptotyping is an iterative, knowledge-based approach that **translates structured research data and curated project knowledge into digital research artefacts** for specific projects.

This process involves **collaboration between LLM-based agents and experts**. New insights are continuously fed back into the knowledge base.



Einführung in Promptotyping. (Teil 1) <https://youtu.be/8sUe4.lkh3uQ>

Einführung in Promptotyping. (Teil 2, Live Demo mit Claude Code). https://youtu.be/hd_a-NBO_S4

Agentic Engineering und digitale Edition mit Claude Code und Fable 5 (Live Demo). <https://youtu.be/kQaTu4oFiSo>

<https://github.com/DigitalHumanitiesCraft/Promptotyping>

Christopher Pollin, 'Promptotyping: Zwischen Vibe Coding, Vibe Research und Context Engineering', L.I.S.A. WISSENSCHAFTSPORTAL GERDA HENKEL STIFTUNG, 17 January 2026, https://lisa.gerda-henkel-stiftung.de/digitale_geschichte_pollin.

AI GENERATED

Hands-On: Document a research perspective

PREPARATION



Data & Sources

Context



EXPLORATION & MAPPING

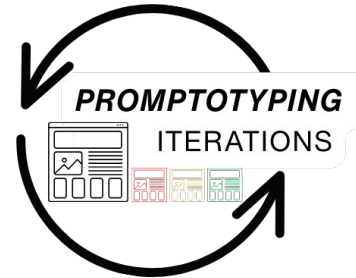
DISTILLATION



DATA.md

REQUIREMENT.md

RESEARCH.md



IMPLEMENTATION

Question

Which publications are exchanged or discussed?

Evidence

Define the relevant source passages and distinctions.

Check

Name a case that would invalidate the proposed result.